

OpenReview export (answer5.tex)

Reviewer A6kZ (guarantees, high- d mechanism, frozen-draw sensitivity)

(1) Convergence guarantees.

Question.

Convergence guarantees. Are there conditions under which convergence is guaranteed for RF-LR models (those mentioned in section 3.5 or otherwise)?

Answer. Our main guarantee is conditional on convergence of the mean-field dynamics. Unconditional convergence rates for deep RF-LR remain largely open; we will clarify this and avoid implying a general unconditional guarantee. We will also point to the sharpest partial results available and clarify which assumptions are sufficient for the global-optimality conclusion given convergence.

(2) High-dimensional extensions of the spike mechanism.

Question.

High-dimension extensions. Does the spike learning mechanism extend to higher dimensions with overlapping spikes in feature space?

Answer. Yes in substance: the mechanism (channel dominance $\leftrightarrow$ amplified backward signal through the mixing matrix and ReLU gates) is dimension-agnostic at the ODE algebra level because it depends on the same identity

$$H_2(c_2; x, W) = \sum_{j=1}^r W_{c_2,j} f_j(x; W)$$

and the channelwise backprop objects that feed the mean-field equations. What we do not currently have for general input dimension d is a clean theorem isolating well-separated (non-overlapping) spikes in feature space; overlapping channel contributions when $d > 1$ complicate bookkeeping and the rigorous two-point caricature used in Theorem 4.1. We will state this distinction clearly in the camera-ready and add empirical evidence in higher d where space permits.

Additional context — Remark after Theorem 4.1 (mechanism beyond the two-point toy; drafted mainly for Reviewer 2 on Theorem 4.1, also cited here because the ODE algebra is dimension-agnostic): The explanatory content is not the simplified two-point, two-channel statement itself, but the positive feedback it isolates. Fix a second-layer neuron index c_2 and recall $H_2(c_2; x, W) = \sum_{j=1}^r W_{c_2,j} f_j(x; W)$. For ReLU, gating enters mean-field gradients through $\varphi'_2(H_2) = \mathbf{1}\{H_2 > 0\}$. The channel- k backward signal (cf. the ODE for $\partial_t w_1(\cdot, k)$) is built from expectations over C_2 of terms that couple (i) the mixing weight $W_{C_2,k}$, (ii) the gate $\mathbf{1}\{H_2(C_2; x, W) > 0\}$, and (iii) upstream factors (output error and later-layer weights). When channel k dominates, neurons c_2 that contribute to the channel- k drift are precisely those with the gate “on.” Across c_2 , the effective weights in the expectation are then positively aligned with $W_{c_2,k}$ and with $\mathbf{1}\{H_2(c_2) > 0\}$, producing a large contribution to the drift of the k -th channel, which in turn pushes f_k further in the same direction: a reinforcing loop. The ODE-level mechanism is dimension-agnostic; the two-point theorem is a minimal tractable instance.

Visualization note. The figures emphasize $d = 1$ mainly for clarity of visualization; the same feedback loop operates when $x \in \mathbb{R}^d$. Overlapping spikes mainly affect how we illustrate and formally isolate regimes, not the core mean-field structure above.

(3) Sensitivity to the frozen feature draw.

Question.

Sensitivity of randomness of fixed feature maps. How sensitive are the convergence guarantees and empirical performance to the initial draw of the frozen feature maps for finite width networks?

Answer. The analytical conditions are not tied to one sampling recipe. Once mixing matrices are frozen, the arguments use an almost-sure channel-aggregation bound (schematically

$\|W^{(\ell)}\|_{\infty,1} \leq rK$) and a richness/diversity assumption for the frozen first-layer features. In principle each frozen $W^{(\ell)}$ can be drawn from any law with finite support (or any law satisfying the same $\ell_{\infty,1}$ control a.s.); there is no Gaussian-specific requirement. In experiments we often use unbounded Gaussian initializations for simplicity at finite width; that is an engineering choice while the theorems use stability/richness predicates. Nonnegative / NMF-style $W \geq 0$ matches the same boundedness template and is intuition for some collapse modes; the ReLU dead-gate caveat is separate (high-probability-in- r only as heuristic; cf. Reviewer VCqo Q10 and the ReLU appendix in the rebuttal PDF). At finite width, performance can vary with the draw. We have not run a systematic post-review sweep over independent frozen-feature draws with optimization held fixed; we will state this in the camera-ready, separate proved assumptions from mean-field idealization, and add compact multi-draw diagnostics if space permits or list them as future work.